Compact PTB case study: same pair-based signal, different aggregation

The compact comparison contrasts k-TSP-10 with the corresponding RLR-10 model on the same ten pair-based contrasts.
In the fitted RLR model, two relations are shrunk to zero and the remaining eight receive unequal weights.

A. Compact k-TSP-10 signature

equal hard voting over all selected relations

Interpretation

Each selected relation contributes the same vote to every prediction.

Selected relation	Aggregation	
ANKRD22 > KLF12		equal vote
PSTPIP2 > IL27RA		equal vote
VAMP5 > ITGAL		equal vote
GBP4 > MAP4K1		equal vote
FCGR1A > ST6GAL1		equal vote
BATF2 > IL1RAP		equal vote
FCGR1BP > HNRNPR		equal vote
GBP5 > LIMS1		equal vote
WARS1 > FCMR		equal vote
GBP6 > ADAM7		equal vote

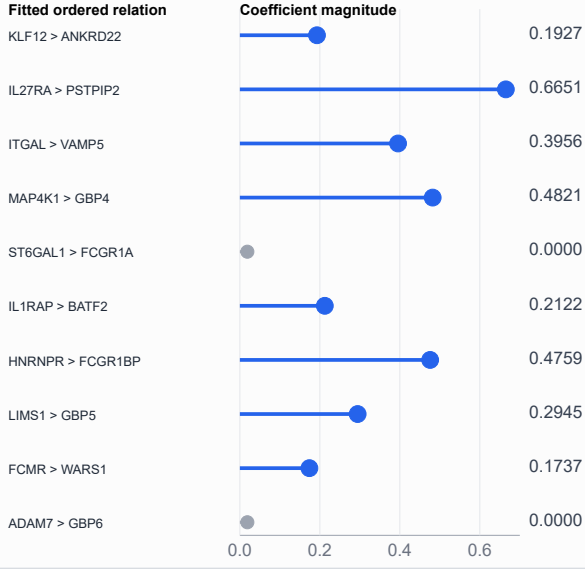
Same pair-based
different aggregation

B. Corresponding RLR-10 fitted coefficients

same biological contrasts; orientation may appear reversed in the fitted export

Interpretation

The pair-based signal is preserved, but relations receive unequal fitted weights.



Local activity summary

Global active pairs = 10; local active pairs per sample = 10 for all test samples.
All selected relations always participate in the final vote.

Local activity summary

Global active pairs = 8; local active pairs per sample = 7 or 8.
Two relations are shrunk to zero and do not enter the fitted weighted core.

Compact PTB case study from the fixed-split evaluation. The matched k-TSP-20 / RLR-20 comparison can be shown analogously in the Supplementary Material.